SmartRouter: Towards Intelligent Search Engine Query Routing

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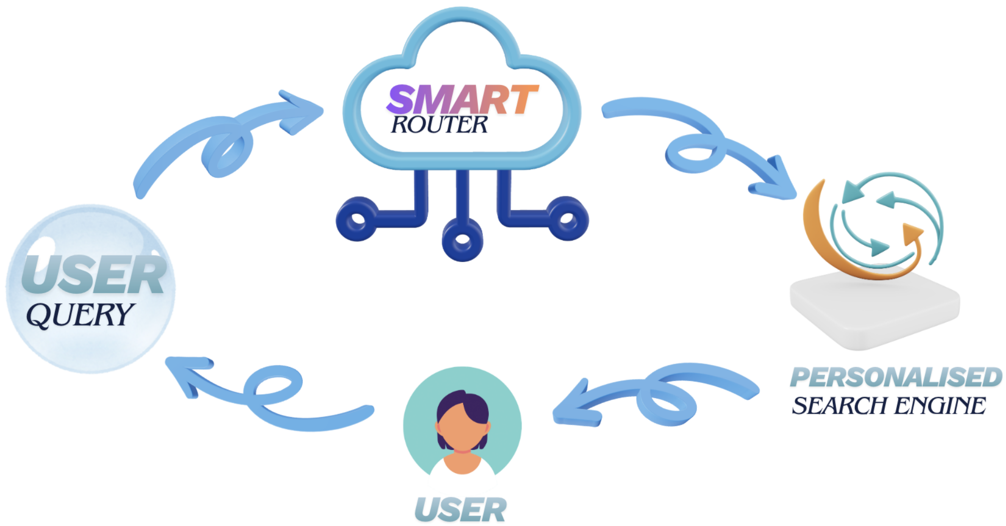

Figure 1: SmartRouter in a nutshell

ABSTRACT

Modern web searching requires users to choose between two types of search engines: traditional search engines like Google or LLM-powered engines like Perplexity. This manual selection introduces friction for the users, often leading to suboptimal results due to satisficing behavior or when a user is unsure which destination best fits a given query. To tackle this problem, we present SmartRouter, a Chrome extension that automatically routes queries to the most appropriate search destination using a lightweight adaptive learning mechanism. Our formative research includes a mixed-methods study combining a mass-market survey (n=52), lab-based usability sessions, and NASA-TLX workload assessments, and semi-structured interviews. We found that SmartRouter reduced effort and cognitive load in users. Our evaluation demonstrates that adaptive routing, paired with minimal-friction interaction design, can streamline search workflows and assist in personalized search behavior over time.

KEYWORDS

Human Computer Interaction, Web Search, Reinforcement Learning, Information Retrieval, Query Intent.

INTRODUCTION

Today's search ecosystem is divided between traditional search that excel at index ranking, and AI-based engines running on LLMs that provide synthesized text and generated code. Despite the foundational nature of these tools, the burden of choosing the correct search destination (SD) still remains on the user. This creates several pain points: users who know their preferred SD endure the hassle of manually navigating between SD tools; users who are unsure which SD would best handle their query default to suboptimal options; and users who are unaware of non-traditional search engines may conclude their query with suboptimal results. The main contributions of SmartRouter are twofold:

(1) An HCI-driven design that integrates user preferences, UI transparency, and override controls without disrupting existing search habits.
(2) A seamless adaptive feedback loop enabling query-level personalization.

RELATED WORK

Understanding user intent has long been central to search and recommendation research. Early work in 2007 by Jansen et al. established foundational taxonomies of search intent [6], while subsequent efforts used deep learning to classify intent at scale. More recent advances leverage user-interaction signals [4, 8] and semantic analysis [2] to detect user goals, resolve ambiguous queries [1], and generate clarifying intent descriptions [13]. LLMs have further unified search and recommendation pipelines [7] by enabling intent-driven session modeling and dual-intent frameworks [12]. Collectively, this body of work reflects a strong emphasis on algorithmic inference within academic settings, in which systems aim to deterministically extrapolate what users want from limited inputs.

In practice, this has manifested in several industry-facing products. The most prominent including;

(1) Google AI Mode and AI Overviews: User-Initiated Workflow and Unsolicited AI-Summaries
Google's recent integration of AI Mode and AI Overviews into its browser is the tech giant's first step in integrating LLM-generated responses directly into the search workflow. Rather than surfacing automatically, AI Mode needs to be triggered intentionally by opening a new tab and pressing Tab + Enter after a query to run it through Google's AI-powered engine. This feature offers a lightweight way for users to switch between traditional and AI-powered search, but still requires them to know which search engine they want to use.
On the other hand, AI Overviews represent the opposite, where LLM-generated summaries appear automatically at the top of the results page. This feature illustrates how Google's predictive systems often override user expectations, as they surface inconsistently, cannot be disabled, and replace traditional indexed pages with synthesized summaries even when users would prefer direct sources.
Together, AI Mode and AI Overviews reflect Google's fragmented approach to intent handling: one relies entirely on user-initiated requests while the other forces model-generated answers on users without user control. This tension reinforces the need for SmartRouter, as users need a system that selects the appropriate SD without requiring shortcuts while remaining fully correctable when prediction misalignment occurs.
(2) E-Commerce Intent Prediction Systems: Adaptive but Non-Correctable
Online commerce platforms like Amazon rely on sophisticated intent-prediction systems to classify user goals, including browsing, purchasing, and comparing products. These systems adapt the interface dynamically to reshape product lists and recommendations based on inferred intent. However, their adaptivity is fundamentally one-directional as the platform predicts and the user must adapt. When the classification misses the mark, users are locked into irrelevant recommendations and have no intuitive way to correct the system's back-end inference.

This mirrors broader issues in the intent-prediction literature, where systems optimize for accuracy but rarely incorporate user-driven feedback loops that update models in real time based on user intention [3, 9]. Our formative research was thus designed to precisely quantify the behavioral and psychological costs incurred by users in this fragmented environment.

FORMATIVE RESEARCH

We employed a Convergent Parallel Mixed-Methods Design to characterize the existing search landscape, focusing on both user behavior and underlying psychological barriers to AI adoption. Our methodology directly addressed the fragmentation identified in the Related Work by seeking to understand how users currently manage the burden of manual destination selection.

We deployed a quantitative survey to 52 participants ($N = 52$) to analyze baseline search behaviors and segment users based on AI receptiveness. The cohort was highly educated, with the majority holding or pursuing advanced degrees. However, a demographic analysis revealed a significant sampling bias toward technical fields, with 59.6% of respondents identifying as "Computer Science / AI / Engineering.". This prompted us to have a more even user panelist in the user testing phase so we can mitigate any bias.

Analysis of search engine rankings revealed a dominant "find, then refine" user model. Over 94% of participants ranked Google as their primary tool for broad discovery, while 69.2% ranked ChatGPT as their secondary tool for synthesis. Notably, familiarity with integrated AI browsers was exceptionally low, with 88.5% of participants indicating they were either unaware of or did not use such tools. This validated the SmartRouter concept as an interaction paradigm for this cohort rather than an iteration on familiar workflows.

To understand psychographic barriers, we constructed an "AI Enthusiasm Score" (Cronbach's $\alpha = 0.87$) based on a 7-item Likert scale. A tertile split revealed three distinct segments ranging from Low to High enthusiasm. To achieve *Verstehen* (understanding user motivations), we analyzed open-ended responses from the "Low Enthusiasm" segment. Three primary barriers to adoption emerged: trust issues regarding "confident wrong answers" (hallucinations), concerns over source attribution and the blending of citations, and fears that over-reliance on AI synthesis would degrade critical thinking skills by creating information "echo chambers." This set of critical user insights specifically prompted us to include Perplexity as the representative LLM-based search engine in our system, as its core design explicitly addresses these concerns by providing the most grounded, citable answers with direct source links.

To enrich survey insights with real-world usage expectations, we engaged with a highly engaged user base by posing our concept to the Perplexity subreddit. Responses from self-identified power-users were illuminating. While many expressed strong enthusiasm for a system that selects the "right engine" automatically, they also insisted on maintaining immediate control to correct system mistakes. One of the upvoted comments captured this sentiment succinctly: *"I would try it but there are many times that I'd like to Google but still type a longer question ... .Would there be an easy one-click switch between them?"* [10]

This feedback directly informed the design of SmartRouter's one-click override mechanism, ensuring that the system accelerates rather than replaces human agency within information seeking.

THE SYSTEM

SmartRouter is a Chrome extension that intercepts user queries typed into the omnibox, and intelligently routes them to the SD most likely to satisfy the user's intent. The current SDs are Google (traditional, index-driven search) and Perplexity (LLM-powered answer engine). The design of SmartRouter reflects a key HCI principle: automation should accelerate the expression of a user's intent, not obscure or replace it.
SmartRouter is implemented entirely on-device using standard Chrome Extensions APIs (omnibox, tabs, and storage modules), with no external services or data collection. The only stored data is

Figure 2: Query entered into SmartRouter Extension

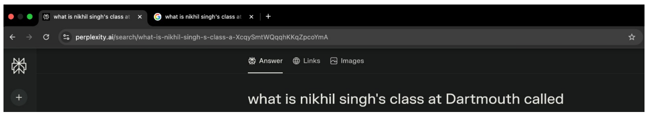

Figure 3: Query from Figure 2 routed to Perplexity, as primary SD

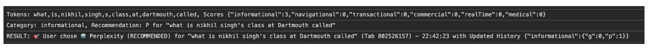

Figure 4: Data records on console logs

a small local preference history, mapping destination choices and corrections, kept in local extension storage to update and improve routing preferences, which users can clear with a single one click. Core Features of SmartRouter:

(1) Automatic Query Routing
SmartRouter uses a lightweight, interpretable classification model, based on query semantics like length, entities, keywords to determine query intent across six categories (e.g informational, navigational, medical, etc). It uses a combination of said global defaults, as well as historical behavior, to make a deterministic guess about where a given query should go. For example, if the user consistently prefers Perplexity for coding queries and Google for navigational or shopping queries, SmarRouter gradually adapts and biases routing toward those patterns.

(2) One-Click Override
A major usability challenge in search workflows is the friction involved in correcting a system's decision-making process. SmartRouter addresses this by opening the alternative SD in the background, allowing users to immediately switch destinations without retyping the query. This solves a core HCI pain point where users know instantly when they choose the wrong SD, but correcting their mistake usually takes multiple clicks. SmartRouter reduces the correction to one intuitive gesture.

(3) Adaptive Learning Feedback Loop
Switching tabs serves as an implicit feedback signal. When a user switches from the system's chosen SD to the alternative, SmartRouter logs this switch locally and updates a Thomson Sampling-inspired heuristic once a small predetermined interaction threshold (set to 7 during the User Studies) is met. Until then, global defaults based on literature review prevents cold-start mistakes. This implements a form of user-correctable personalization, in which the model improves not through opaque inference but through iteration, which preserves user agency.

(4) Background Tab Auto-Close
For all queries, SmartRouter routes the current tab to the recommended SD, and the alternative SD in a separate background tab. To prevent clutter and preserve tab organization, a significant pain point identified in Stage 2 of our user study, the system automatically closes the unused SD after a short delay (set to 11 seconds for the prototype). This maintains UI orderliness while still providing users a confidence window to reinterpret the intent of a given query. Future versions will scale the timeout by category dynamically to enable a more seamless and clutterless searching experience.

EVALUATION

To rigorously assess the impact of SmartRouter on search efficiency and user experience, we conducted a quantitative evaluation focusing on two primary dimensions: Cognitive Load (via NASA-TLX) and Information Scent (via Behavioral Override Rates). This evaluation employed a within-subjects design ($N = 4$), where participants performed information retrieval tasks under both Baseline (Control) and Treatment (SmartRouter) conditions.

Experiment 1: Cognitive Load Analysis

We hypothesized that SmartRouter would reduce the cognitive burden associated with selecting search destinations. To measure this, we administered the NASA-TLX assessment [5] following each condition. We analyzed the results using paired-sample t-tests and calculated Cohen's d to determine effect sizes.

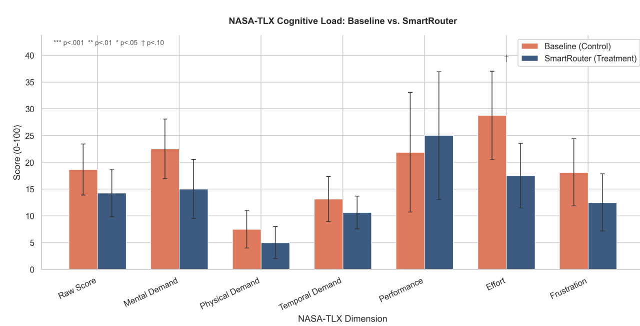

Figure 5: Comparison of mean NASA-TLX scores. SmartRouter (Blue) shows consistently lower load than Baseline (Orange).

Results. Our analysis of the NASA-TLX metrics reveals a consistent reduction in cognitive load when using SmartRouter. As illustrated in Figure 6, participants reported lower scores across 6 of the 7 dimensions in the treatment condition compared to the baseline.

The most substantial impact was observed in the Effort dimension. Figure 5 visualizes the effect sizes, highlighting that SmartRouter reduced perceived effort by nearly 39% ($p = 0.065$). This large effect size ($d = -0.77$) suggests the system successfully offloaded the cognitive "heavy lifting" of destination selection. Reductions were also evident in Mental Demand (33% decrease) and Frustration

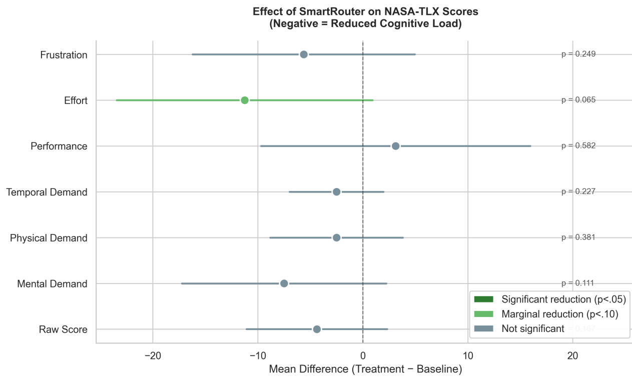

Figure 6: Forest plot of effect sizes. The shift to the left indicates a reduction in cognitive load, with *Effort* showing the strongest effect.

(31% decrease), reinforcing the conclusion that the system made the search process feel significantly less taxing.

Table 1 summarizes the paired t-test results for these key dimensions.

Table 1: NASA-TLX Paired t-Test Results (Baseline vs. Treatment)

Metric	Base (M)	Treat (M)	Δ	$t(7)$	p	Cohen's d
Raw Score	18.65	14.27	-4.38	1.54	.167	-0.55
Mental Demand	22.50	15.00	-7.50	1.82	.111	-0.64
Effort	**28.75**	**17.50**	**-11.25**	**2.18**	**.065**	**-0.77**
Frustration	18.12	12.50	-5.62	1.26	.249	-0.44
Performance	21.88	25.00	+3.12	-0.58	.583	+0.20

Experiment 2: Behavioral Override and Trust

To objectively measure user trust, we tracked the Override Rate, instances where a user actively rejected the system's choice. Drawing on Information Foraging Theory [11], a low override rate signifies a strong "Information Scent," implying that the system correctly anticipated the user's navigational goal.

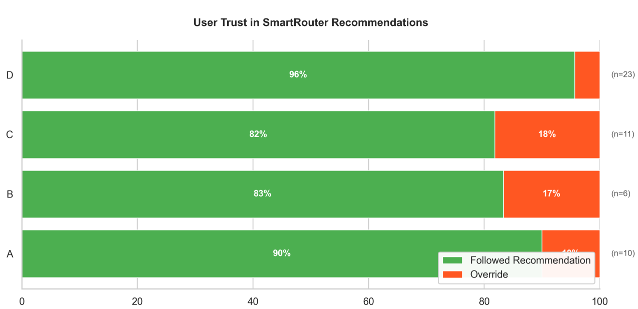

Figure 7: User Trust Breakdown. The dominance of green indicates that users accepted the system's routing decision in 90% of all queries.

Results. Our behavioral logs reveal a decisive alignment between system predictions and user intent. As illustrated in Figure 7, users accepted the SmartRouter's recommendation in the overwhelming majority of cases (90%), rarely feeling the need to intervene.

Figure 8 further breaks down this behavior by participant. Notably, we observed a positive correlation between usage volume and trust; the participant with the highest query count demonstrated the lowest override rate. This suggests that as users became more familiar with the system, they felt comfortable relinquishing manual control, effectively validating the system's ability to support efficient information foraging without friction.

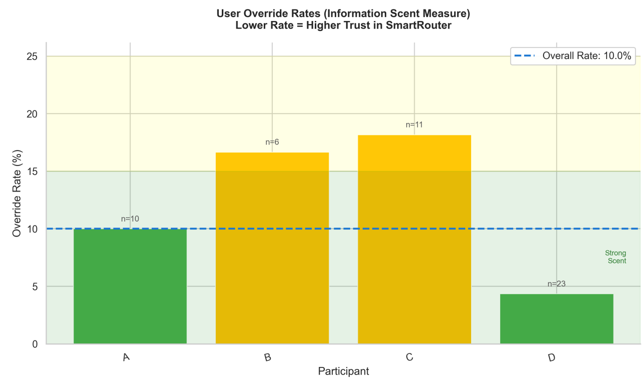

Figure 8: Individual Override Rates. The dashed line represents the 10% aggregate average. Lower bars indicate higher trust/stronger information scent.

Qualitative Interviews

Semi-structured interviews provided insight into how users experienced SmartRouter's routing recommendations. Overall, participants reported strong utility in being relieved of the decision-making burden of selecting a SD. Many noted that determining the "right" site, especially when queries required domain-specific webpage sources, often feels like unnecessary cognitive work, and SmartRouter effectively removed that friction.

Despite this, participants identified interaction pain points. The aforementioned automatic timeout before tab switching was consistently described as too short, limiting their ability to process initial results. Users requested a longer delay or a mechanism to easily restore closed tabs, emphasizing the need to maintain user control.

Participants generally perceived the category system as well-structured and aligned with their expectations. One participant who executed a wide variety of self-initiated queries appreciated SmartRouter's consistent choices of default SDs. However, others surfaced edge cases where personal preferences diverged from the system's learned defaults. For instance, one technically experienced user preferred coding-related queries to route to Google/StackOverflow, whereas SmartRouter classified them as Informational and sent them to Perplexity. Conversely, another user found that behavior ideal. This variability motivated exploration of user-customizable subcategories (e.g., "Coding" under Informational) to allow SmartRouter to adapt to individual search behavior without degrading routing accuracy for other intent types.

DISCUSSION

The convergent evidence from both experiments suggests that SmartRouter effectively bridges the gap between user intent and search destination.

The data shows a clear link between the behavioral and psychometric findings. The 90% acceptance rate in Experiment 2 explains the 39% reduction in perceived *Effort* in Experiment 1. By accurately predicting the correct search destination, the system removed the micro-decision friction typically involved in search, directly lowering the cognitive cost of the task. Furthermore, the reduction in *Frustration* ($d = -0.44$) correlates with the high reliability of the routing; users rarely encountered "dead ends" that required manual correction.

While the effect sizes (Cohen's d) ranged from medium to large, the statistical significance was limited by the small sample size ($N = 4$). However, the consistent directionality of the data—reduced load and high trust across all participants—provides strong preliminary suggestion for the system's efficacy. Future work will focus on scaling the participant pool to achieve the statistical power necessary ($n \approx 34$) to confirm these effects at the $\alpha < .05$ level.

Qualitative feedback further reinforced SmartRouter's value in reducing the "micro-decision" burden during search, leading to a smoother and more efficient user experience. At the same time, participants surfaced personalized friction points, most notably the need for customizable sub-categories and a more adaptive background-tab close timer, which now clearly shape our priorities for future design iterations and an intended public release later this year.

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